E. Number With The Given Amount Of Divisors

time limit per test: 2 seconds
memory limit per test: 256 megabytes

Given the number n, find the smallest positive integer which has exactly n divisors. It is guaranteed that for the given n the answer will not exceed 10^{18}.

Input

The first line of the input contains integer n ($1 \le n \le 1000$).

Output

Output the smallest positive integer with exactly n divisors.

Examples

input Copy

```
4
```

output Copy

```
6
```

input Copy

```
6
```

output Copy

```
12
```

→ Attention

The package for this problem was not updated by the problem writer or Codeforces administration after we've upgraded the judging servers. To adjust the time limit constraint, a solution execution time will be multiplied by 2. For example, if your solution works for 400 ms on judging servers, then the value 800 ms will be displayed and used to determine the verdict.

Codeforces Beta Round 27 (Codeforces format, Div. 2)

Finished

Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, msys2

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
372427924	Apr/25/2026 10:11	Accepted
372427781	Apr/25/2026 10:10	Accepted
372427566	Apr/25/2026 10:08	Wrong answer on test 2

372427426	Apr/25/2026 10:07	Compilation error
372427306	Apr/25/2026 10:06	Compilation error
372427169	Apr/25/2026 10:05	**Accepted**
372426937	Apr/25/2026 10:03	Wrong answer on test 2
372426719	Apr/25/2026 10:01	Wrong answer on test 2
372423375	Apr/25/2026 09:59	Compilation error
372423181	Apr/25/2026 09:57	Wrong answer on test 2

→ Problem tags

brute force | dp | number theory | *2000

No tag edit access

→ Contest materials

- Announcement
- Tutorial #1
- Tutorial #2 (en)
- Tutorial #3 (en)

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